Team Name : Starcy

Team Leader Name : Piyush Rajendra Yenorkar

Problem Statement : Infrared Image Colorization and Enhancement for Improved Object Interpretation.

Team Members

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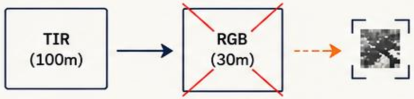
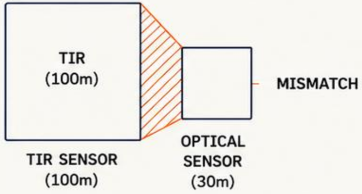
SPECTRA solves the resolution mismatch by explicitly decoupling enhancement and colorization.

The Flaw in Existing Ideas

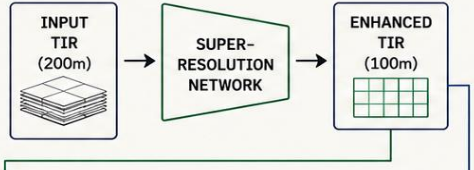
- Existing methods treat IR colorization as a single mapping problem.
- **The Mismatch:** They ignore the fundamental hardware reality—TIR sensors capture at 100m native resolution, while optical sensors capture at 30m. Single-stage models fail here.



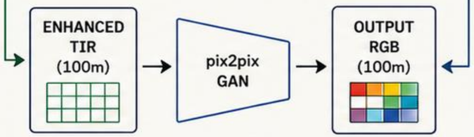


The SPECTRA Solution

- SPECTRA is the first approach to explicitly decouple the problem into a dedicated two-stage cascade pipeline.
- **Stage 1: Super-Resolution Sub-Pixel Network** (upscales the 200m TIR to 100m, recovering spatial details).

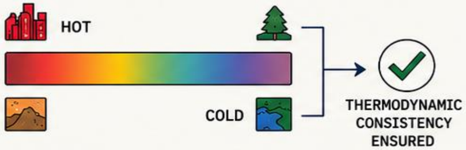


- **Stage 2: pix2pix GAN** maps the enhanced 100m TIR to 100m RGB.



The Winning USP

- **Physics-Informed Modeling:** A custom thermal emissivity constraint ensures thermodynamic consistency (e.g., hot pixels must map to urban/bare soil colors; cold pixels to vegetation/water).



- This explicitly **eliminates** thermodynamically impossible color assignments, directly preventing hallucinations as requested in ISRO's evaluation criteria.

HALLUCINATION
(IMPOSSIBLE ASSIGNMENT)

SPECTRA OUTPUT
(PHYSICALLY VALID)

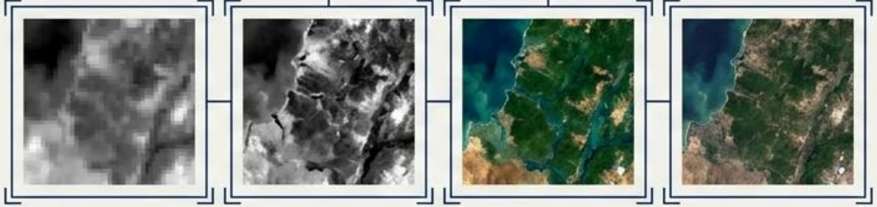
An end-to-end trained Deep Learning pipeline delivering precise enhancement and colorization.

Core ML Pipeline

- Two-stage architecture (SpectraSatNet) with end-to-end shared feature representations.
- Bonus physics-informed thermal emissivity loss.

Evaluation Framework

- Automated quantitative reporting for PSNR, SSIM, and FID.



1: 200m TIR Input

2: 100m SR Output

3: Colorized RGB

4: Ground Truth

Data Pipeline

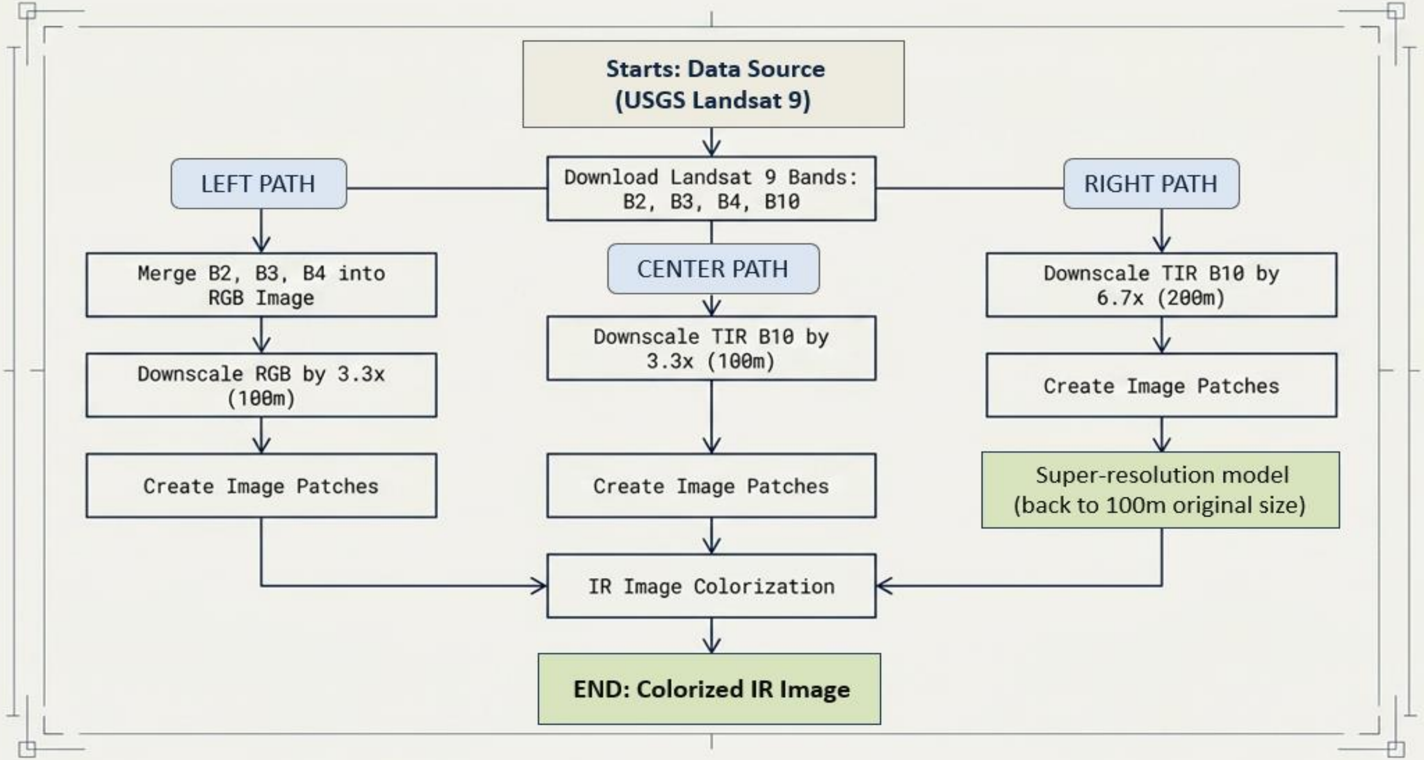
- Automated Landsat 9 TIR1-RGB pairs acquisition spanning 8 geographically diverse Indian terrain categories.
- B2, B3, B4, B10 bands processed.

Deployment

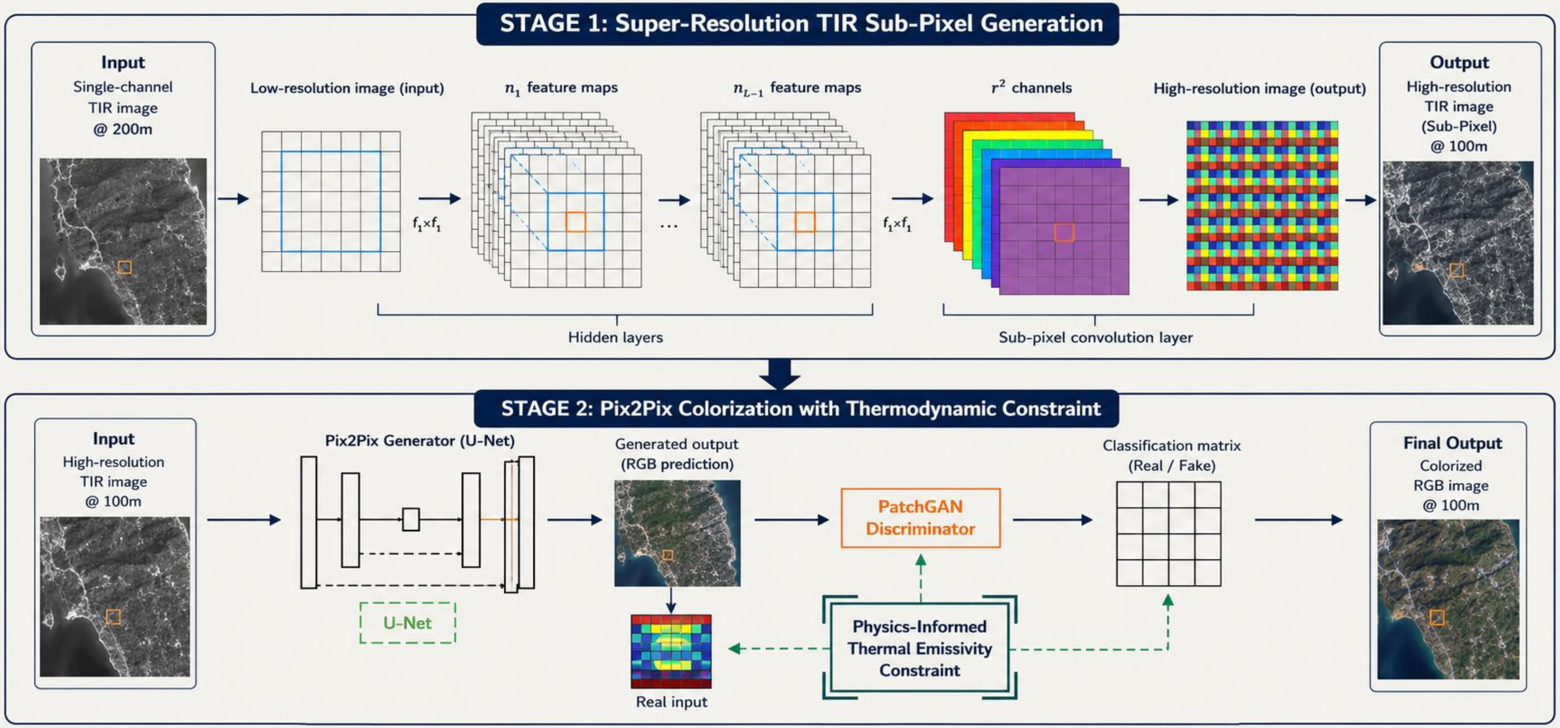
- Low inference time per tile via ONNX model export.

Validation Note: We have validated our preprocessing pipeline and model architecture on Landsat 9 data. Full training will be completed using ISRO's official dataset instructions upon release.

Processing Landsat 9 TIR1-RGB pairs precisely to ISRO specifications.



A unified architecture constrained by thermodynamic reality to prevent hallucinations.



A fully open-source, Python-based end-to-end solution optimized for scale.

Deep Learning Models

- PyTorch 2.0+
- pix2pix GAN (U-Net + PatchGAN)
- Sub-Pixel Convolution (PixelShuffle)
- Custom Thermal Emissivity Loss

Data Handling & I/O

Data I/O

- Google Earth Engine API (Automated Landsat 9 download)
- Rasterio
- GDAL
- tifffile, OpenCV

Evaluation & Diagnostics

- PSNR
- SSIM (scikit-image)
- clean-fid (dataset-level evaluation)
- Custom Visual Inspection Module

Compute & Deployment

- ONNX Runtime (Low inference time per tile)
- Google Colab (T4 GPU training infrastructure)

Enterprise-grade thermal modeling achieved entirely with zero-cost infrastructure.

Component	Cost	Description & Notes
Landsat 9 Level-2 Data (USGS)	Free	Accessed via Google Earth Engine API.
Google Colab Compute (T4 GPU)	Free	Sufficient for complete model training (~6-8 hours).
Python + PyTorch + Libraries	Free	Entirely open-source ecosystem.
ONNX Runtime Inference Engine	Free	Open-source deployment for low-latency per-tile processing.
Total Estimated Cost	₹0	No proprietary tools or paid datasets required.

"Working Codebase & Architecture Preview: [https://github.com/piyushyenorkar/spectraisro]"

Interactive visual inspection module for real-time hallucination detection.

Target: Inference <50ms/tile | PSNR >28 dB | SSIM >0.85 | FID (dataset-level)

- Designed specifically for qualitative visual inspection to prevent hallucinations.
- Optional lightweight deployment for side-by-side comparison of IR Input vs. Enhanced Output vs. Reference Imagery.
- FID (Fréchet Inception Distance) computed at dataset level across full validation set — not per-tile.

BHARATIYA ANTARIKSH HACKATHON 2026

THANK YOU